A virtual class of nonconforming finite elements and its applications to Poisson's equation

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Abstract

In this paper, we develop and analyze the virtual class of nonconforming Finite Element inspired from the classical Crouzeix-Raviart element in two and three-dimensions. We focus on Poisson's equation for theoretical and numerical results. We show that the discrete problem is stable and the *a priori* error estimates are optimal. Two numerical tests are presented to demonstrate the theoretical results.

Keywords: Nonconforming finite element, element of Crouzeix-Raviart, interpolation error, consistency error, *a priori* error estimates.

1. Introduction

The nonconforming finite element methods are very popular method for the numerical approximation of partial differential equations, for they enjoy better stability properties and smaller support compared to the conforming finite element methods. Moreover, it is easier to prove the inf-sup condition for Stokes equations and it is also quite simple to implement. One of the most popular non- conforming finite element methods is the method based on discretizing with the lowest order Crouzeix-Raviart element ([5], see [8] for many improvements and generalizations). It is also known as simplicial P^1 nonconforming finite element which expression is piecewise linear polynomials with respect to the triangulation of domain while degrees of freedom are the mean of the function over the edges/facets ($2D/3D$). They interested many mathematicians and numericians with different applications, see for example [1, 3, 4, 6, 10, 12, 16, 17].

Our interest in the topic of this work arose from the fact that there exist original papers which use and discuss the development of the Crouzeix-Raviart element, see for instance, [7, 11, 13–15], but none of them has proposed a new finite element with the degrees of freedom are the vertices of the simplex used for Crouzeix-Ravairt element. Therefore, the main of this paper is to carry out

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the development and the numerical analysis of a new class nonconforming finite element inspired by the classical Crouzeix-Raviart.

The rest of this paper is organized as follows: First, we introduce and develop the virtual family of Crouzeix-Raviart finite element. Then, in the third section, we introduce the useful notations, Poisson's equation and its associated weak formulation by considering the new triangular/tetrahedral nonconforming finite element. In the fourth, we establish the estimate of interpolation error for our finite element. Then, in the fifth section, we interest in consistency error and we establish the results proving the optimal convergence of *a priori* error estimates. In the final section of this paper we present two numerical experiments to demonstrate the theoretical results and we draw the conclusion.

Throughout the paper we consider the following convention concerning a bound constant. We will note by c a positive constant independent of the indices and the simplex. Thus, we do not need to precise its expression and it may vary from one line to the other, and even in the same line.

2. Virtual class of nonconforming finite element

Let T be non-degenerate simplex in $\mathbb{R}^d$, $d = 2, 3$ that composed of affine d-simplices (i.e. triangles if $d = 2$ or tetrahedrons if $d = 3$) with vertices v_i, $i \in \{1, \cdots, d+1\}$ and d-dimensional Lebesgue measure $|T|$. By λ_i, $i \in \{1, \cdots, d+1\}$, we denote the barycentric coordinates for a point v_i, $i \in \{1, \cdots, d+1\}$, relative to a simplex T. Moreover, we introduce F_i, $i \in \{1, \cdots, d+1\}$ the facets of T such that F_i means an edge in $\mathbb{R}^2$ and face in $\mathbb{R}^3$, but this distinction will not be made unless necessary. Finally, we introduce the simplex $\tilde{T}$ which contains T with vertices $\tilde{v}_i$, $i \in \{1, \cdots, d+1\}$. To ease the notation, we occasionally refer to $T \subset \mathbb{R}^d$ and $\tilde{T} \subset \mathbb{R}^d$, see Figure 1.

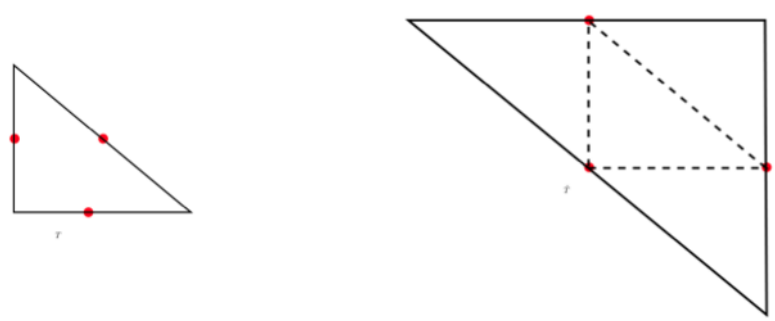

Figure 1: Illustration in two dimension of Crouzeix-Raviart finite element on T (left) and our finite element on $\tilde{T}$ (right).

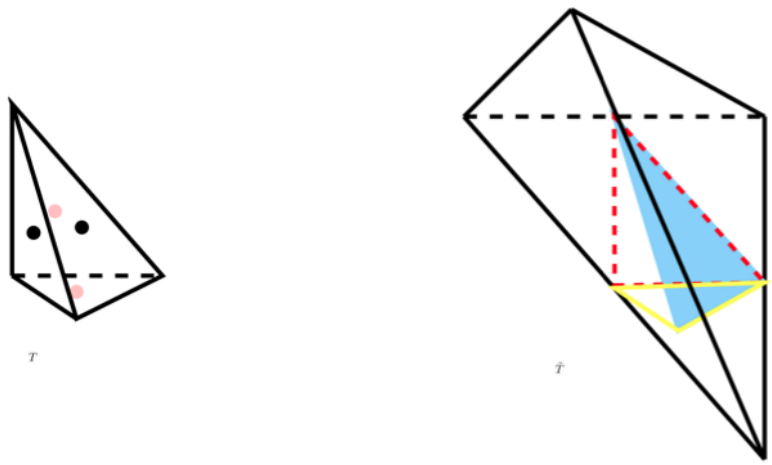

Figure 2: Illustration in three dimension of Crouzeix-Raviart finite element on T (left) and our finite element on $\tilde{T}$ (right).

To develop our finite element, we first introduce briefly the d-dimensional Crouzeix-Raviart element on T; $\left(T, P^1(T), \Sigma^{CR}\right)$ and it is expressed by:

1. $P^1(T)$ is the space of polynomials of degree 1 on T.
2. The degrees of freedom of Crouzeix-Raviart are

$$\Sigma_T^{CR} := \{\sigma_i^{CR},\, i = 1, 2, \cdots, d+1\} \tag{2.1}$$

with

$$\sigma_i^{CR} := \frac{1}{meas(F_i)} \int_{F_i} v ds, \quad i = 1, 2, \cdots, d+1. \tag{2.2}$$

In addition, the basis functions in terms of the barycentric coordinates of the edge/face, see [9], are expressed in the form:

$$\varphi_i^{CR}(x) = d\left(\frac{1}{d} - \lambda_i(x)\right), \quad i = 1, 2, \cdots, d+1. \tag{2.3}$$

In this work, we interest in development of a new finite element in the simplex $\tilde{T}$ (see Figures 1 and 2). To do this, we introduce the notation of vertices $\tilde{v}_i$ on $\tilde{T}$ such that

$$\tilde{v}_i = \sum_{j=1, j\neq i}^{d+1} v_j + (1-d)v_i \tag{2.4}$$

Hence, (2.4) can be rewritten as follows:

$$\tilde{v}_i = \sum_{j=1}^{d+1} v_j + dv_i \tag{2.5}$$

By using the above notations, we cite the following lemma which is the key to development of our finite element.

Lemma 2.1. *Let $\tilde{T}$ be a simplex with vertices $\tilde{v}_i$, $i = 1, \cdots, d+1$. Then $\tilde{T}$ is non-degenerate with vertices $\tilde{v}_i$, $i = 1, \cdots, d+1$*

Proof. By using (2.5), we have

$$\begin{aligned}\tilde{v}_i - \tilde{v}_1 &= \sum_{j=1}^{d+1} v_j + dv_i - \left(\sum_{j=1}^{d+1} v_j + dv_1\right) \qquad = -dv_i + dv_1 \\ &= -d(v_i - v_1)\end{aligned}$$

Thus, by multiplying both sides of the last equation by α_i and summing for $i = 2$ to $(d+1)$, we obtain

$$\sum_{i=2}^{d+1} \alpha_i(\tilde{v}_i - \tilde{v}_1) = -d \sum_{i=2}^{d+1} \alpha_i(v_i - v_1) \tag{2.6}$$

Finally, from the above equation (2.6), we have

$$\alpha_i = 0, \quad i = 2, \cdots, d+1. \tag{2.7}$$

Here, we use that the simplex T is non-degenerate, then the elements of $\{v_i - v_1,\ i = 2, \cdots, d+1\}$ are linearly independent.

Now, we present the lemma which introduces the barycentric coordinates in a simplex $\tilde{T}$.

Lemma 2.2. *The barycentric coordinates associated to $\tilde{v}_i$, $i = 1, \cdots, d+1$ are $\tilde{\lambda}_i$, $i = 1, \cdots, d+1$, such that*

$$\tilde{\lambda}_i = \frac{1 - \lambda_i}{d}. \tag{2.8}$$

Proof. It suffices to show that for all $x \in \tilde{T}$:

$$x = \sum_{i=1}^{d+1} \frac{1 - \lambda_i}{d} \tilde{v}_i, \quad \sum_{i=1}^{d+1} \tilde{\lambda}_i = 1. \tag{2.9}$$

Indeed, we have

$$\begin{aligned}\sum_{i=1}^{d+1} \frac{1 - \lambda_i}{d} \tilde{v}_i &= \sum_{i=1}^{d+1} \frac{1 - \lambda_i}{d} \left(\sum_{j=1}^{d+1} v_j - dv_i\right) \\ &= \frac{1}{d}\left[\left(\sum_{j=1}^{d+1} v_j\right) \sum_{i=1}^{d+1} (1 - \lambda_i) - d \sum_{i=1}^{d+1} (1 - \lambda_i) v_i\right] \\ &= \sum_{j=1}^{d+1} v_j - \sum_{i=1}^{d+1} v_i + \sum_{i=1}^{d+1} \lambda_i v_i = x, \quad \forall x \in \tilde{T}.\end{aligned}$$

Then we get

$$\sum_{i=1}^{d+1} \tilde{\lambda}_i \tilde{v}_i = \sum_{i=1}^{d+1} \lambda_i v_i, \quad \forall x \in \tilde{T}. \tag{2.10}$$

By using $x = \sum_{i=1}^{d+1} \lambda_i v_i$, $\forall x \in \tilde{T}$ that is always true in $\mathbb{R}^d$, (2.10) reads

$$\sum_{i=1}^{d+1} \tilde{\lambda}_i \tilde{v}_i = x. \tag{2.11}$$

Moreover, it easy to prove the following equality:

$$\sum_{i=1}^{d+1} \tilde{\lambda}_i(x) = \sum_{i=1}^{d+1} \frac{1-\lambda_i(x)}{d} = 1. \tag{2.12}$$

Finally, we prove that the coefficients $\tilde{\lambda}_i$, $i = 1, \cdots, d+1$, are the barycentric coordinates in $\tilde{T}$ associated to $\tilde{v}_i$, $i = 1, \cdots, d+1$.

By using the result of Lemma 2.2, we introduce the lemma of the non-degeneracy.

Lemma 2.3. *The non-degenerate simplex T is a subset of the large simplex $\tilde{T}$. Then, we write $T \subset \tilde{T}$.*

Proof. It is easy to prove this lemma. Let us $x \in T$, then we have

$$x = \sum_{i=1}^{d+1} \lambda_i v_i = \sum_{i=1}^{d+1} \tilde{\lambda}_i \tilde{v}_i = . \tag{2.13}$$

For all $i = 1, \cdots, d+1$, let $\mathrm{F}\tilde{F}_i$ be the face/edge of $\tilde{T}_i$ such that $\tilde{\lambda}_i = 0$ ($\lambda = 1$). Then, Lemma 2.3 combined by the arguments used in its proof gives the following result.

Lemma 2.4. *For all $i = 1, \cdots, d+1$, we have*

(i) $x \in \tilde{F}_i \cap T \Longleftrightarrow x = v_i$.

(ii) v_i *is the center of gravity of the face/edge* $\tilde{F}_i$.

Proof. Let $v_i \in P(T)$ then

$$v_i = \sum_{i=1}^{d+1} \lambda_j(v_i) v_j, \tag{2.14}$$

where the barycentric coordinates associated to v_i, $i = 1, \cdots, d+1$ are

$$\lambda_j(v_i) \begin{cases} 1 & if\, i = j, \\ 0 & if\, j \neq i. \end{cases} \tag{2.15}$$

Then, we get

$$\tilde{\lambda}_j(v_i) = \frac{1-\delta_{ij}}{d}, \tag{2.16}$$

where δ_{ij} stands for the Kronecker delta function. As $v_i \in \tilde{T}$ and by the property (i) of the Lemma 2.4, it follows that

$$v_i = \sum_{j=1}^{d+1} \lambda_j(v_i)\tilde{v}_j = \sum_{j=1}^{d+1} \frac{1-\delta_{ij}}{d}\tilde{v}_j = \frac{1}{d}\sum_{i=1, j\neq i}^{d+1} \tilde{v}_j \tag{2.17}$$

Therefore, the required result holds.
We now prove that (ii) implies (i). Let $x = v_i$, we have

$$v_i = \sum_{j=1}^{d+1} \lambda_j(v_i)v_j = \sum_{j=1}^{d+1} \tilde{\lambda}_j(v_i)\tilde{v}_j \tag{2.18}$$

Since, $\lambda_j(v_i) = \delta_{ij}$, one can deduce, from (2.15) and (2.16), that

$$v_i = \frac{1}{d}\sum_{j=1, j\neq i}^{d+1} \tilde{v}_j. \tag{2.19}$$

Then, we have $x \in \tilde{F}_i \cap T$, as desired. This concludes the proof.

In order to determine the basis functions $(\tilde{\varphi}_i)_{1\leqslant i\leqslant d+1}$ associated to the edges/facets of $\tilde{T}$, it is now sufficient to replace λ_i in (2.3) by $1 - d\lambda_i$, which yields:

$$\tilde{\varphi}_i(\mathbf{x}) = 1 - d - d^2\tilde{\lambda}_i(\mathbf{x}). \tag{2.20}$$

Remark 2.5. *We remark that the basis functions $(\tilde{\varphi}_i)_{1\leqslant i\leqslant d+1}$ associated to the edges/facets of $\tilde{T}$ satisfy the following relation:*

$$\sum_{i=1}^{d+1} \tilde{\varphi}_i = \sum_{i=1}^{d+1} \varphi_i^{CR} = 1. \tag{2.21}$$

We are now able to conclude that $(\tilde{\varphi}_i)_{1\leqslant i\leqslant d+1}$, is a basis of the space of polynomials of degree 1 on $\tilde{T}$; $P^1(\tilde{T})$. Therefore, we summarize and state below the following theorem, which characterizes our nonconforming finite element.

Theorem 2.6. *Let $\tilde{T}$ be a simplex in $\mathbb{R}^d$, $P^1(\tilde{T})$ be the space of polynomials of degree 1 on $\tilde{T}$, and define the degrees of freedom to be the values over the $(d+1)$ facets/edges F_i of T, i.e.,*

$$\tilde{\sigma}_i = v(F_i), \quad \forall i = 1, \cdots, d+1. \tag{2.22}$$

Let us $\tilde{\Sigma} = (\tilde{\sigma}_i)_{1\leqslant i\leqslant d+1}$. Then, the triplet $\left(\tilde{T},\, P^1(\tilde{T}), \tilde{\Sigma}\right)$ is a finite element.

Now we are in the position to estimate the H^1 semi-norm and the L^2-norm of the basis functions $(\tilde{\varphi}_i)_{1\leqslant i\leqslant d+1}$.

Proposition 2.7. *Let $\tilde{T}$ be a simplex in $\mathbb{R}^d$. Then the following estimates*

$$\sum_{i=1}^{d+1} |\tilde{\varphi}_i|_{1,\tilde{T}} \leqslant cd^2, \quad \sum_{i=1}^{d+1} \|\tilde{\varphi}_i\|_{0,\tilde{T}} \leqslant cd^2\sqrt{|\tilde{T}|} \tag{2.23}$$

hold, where a constant $c > 0$ is independent of $\tilde{T}$.

Proof. By using (2.20) and $\nabla\tilde{\lambda}_i$ is constant, it follows that

$$\sum_{i=1}^{d+1} |\tilde{\varphi}_i|_{1,\tilde{T}} \leqslant cd^2, \tag{2.24}$$

this leads to the first estimate of Proposition 2.7 in H^1 semi-norm.
We are now ready to prove the second estimate in L^2-norm. By using (2.20) and passing to the reference element, we get

$$\sum_{i=1}^{d+1} \|\tilde{\varphi}_i\|_{0,\tilde{T}} \leqslant cd^2\sqrt{|\tilde{T}|} \tag{2.25}$$

In order to make the lecture more fluent, we drop tilde symbol in the notation: for example, we write only v, T instead of $\tilde{v}$ and $\tilde{T}$, respectively.

3. A Prior Discretization Error Estimates

Before we state the nonconforming finite element method, we first introduce some notation: Ω denotes d-dimensional bounded open set with polygonal boundary $\partial\Omega$ and $(\mathcal{T}_h)$ is regular family of triangulation of Ω, each $\mathcal{T}_h$ consisting of simplices of diameter $h_T := |T|^{1/2}$ and we set $h = \max\limits_{T\in\mathcal{T}_h} h_T$. Furthermore, let $\mathcal{F}_h$ be the set of sides of the triangulation $\mathcal{T}_h$, while $\mathcal{F}^{int}$ denotes the set of h sides contained in Ω and $\mathcal{F}^{\partial}$ the set of sides situated on the boundary $\partial\Omega$. For $F \in \mathcal{F}_h$, we, in addition, fix once for all a unit normal n_F , defines the elements $T^+ \in \mathcal{T}_h$ and $T^- \in \mathcal{T}_h$, with $F = T^+ \cap T^-$; if F is located on $\partial\Omega$, then n_F is the outward normal n_Ω.

Finally, for $x \in F$ and v a piece-wise smooth function, we define

$$v_h^+(x) := \lim_{\varepsilon\longrightarrow 0} v(x - \varepsilon n_F), \quad v_h^-(x) := \lim_{\varepsilon\longrightarrow 0} v(x + \varepsilon n_F) \tag{3.1}$$

and its jump across and weighted means in $F \in \mathcal{F}_h$ are given as follows:

$$[v]_F := \begin{cases} v_h^+ - v_h^- & if\, F \in \mathcal{F}^{int} \\ v_h & if\, F \in \mathcal{F}^{\partial} \end{cases}, \{v\}_F := \frac{1}{2}\left(v_h^+ + v_h^-\right). \tag{3.2}$$

We now consider the model problem of solving $u \in H_0^1(\Omega)$ such that

$$-(\Delta u, \Delta v) = (f, v), \quad \forall v \in H_0^1(\Omega) \tag{3.3}$$

where the right hand side $f \in L^2(\Omega)$ is a given source term and $(\cdot,\cdot)$ is the usual L^2-inner product. On Th, we introduce the finite element space

$$V_h := \left\{ v_h \in L^2(\Omega) : \, v_h \in P^1(T) \, \forall T, \int_F [v]_F d\sigma = 0 \, \forall F \in \mathcal{F}^{int} \cup \mathcal{F}^{\partial} \right\}, \tag{3.4}$$

equipped with H^1-norm: $\|\cdot\|_{1,\Omega}^2 = |\cdot|_{1,\Omega}^2 + \|\cdot\|_{0,\Omega}^2$ where

$$|\cdot|_{1,\Omega}^2 = \sum_{T \in \mathcal{T}_h} |\cdot|_{1,T}^2, \quad \|\cdot\|_{0,\Omega}^2 = \sum_{T \in \mathcal{T}_h} \|\cdot\|_{1,T}^2 \tag{3.5}$$

We note that $V_h \nsubseteq H_0^1(\Omega)$, that means the method is nonconforming. Thus, ∇u_h is not defined on inter-element boundaries and the variational formula of (3.3) is: finding $u_h \in V_h$ such that

$$a_h(u_h, v_h) = l_h(v_h), \quad \forall v_h \in V_h \tag{3.6}$$

with

$$a_h(u_h, v_h) := \sum_{T \in \mathcal{T}_h} \int_T \nabla u_h \cdot \nabla v_h \, d\mathbf{x},$$
$$l_h(v_h) := \sum_{T \in \mathcal{T}_h} \int_T f v_h \, d\mathbf{x}.$$

The uniform V_h-stability of $a_h(\cdot,\cdot)$ is completely standard. Then, it gives the well-posedness in the variational formula (3.6). In addition, the *a priori* error estimate satisfies the Strang Lemma [6]:

$$\|u - u_h\|_{1,\Omega} \leqslant c \left(\inf_{v_h \in V_h} \|u - v_h\|_{1,\Omega} + \sup_{w_h \in V_h, w_h \neq 0} \frac{|a_h(u - u_h, w_h)|}{\|w_h\|_{1,\Omega}} \right) \tag{3.7}$$

where the constant c is independent of h. In the sequel, we will be interested in estimating of the interpolation error and the consistency error estimate of (3.7). The first one is estimated by using $v_h = I_h v$ with the suitably defined interpolation operator I_h, see the section 4 A second one is treated by the Galerkin orthogonality described in section 5.

4. Interpolation error estimates

In this section we establish the interpolation error estimates for both two and three dimensions under considerations. For our finite element $\big(T, P^1(T), \sigma\big)$, we denote by I_T the local interpolation operator defined from $H^s(T)$ to $P^1(T)$

with $s = 1, 2$, by means of the previous degrees of freedom, associated to the edges/faces such that:

$$I_T := \sum_{i=1}^{d+1} \left(\frac{1}{meas(F_i)} \int_{F_i} v \, d\sigma \right) \varphi(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^d \tag{4.1}$$

where φ, $i = 1, \cdots, d+1$, are given by .

Now, we are in the position to define the global interpolation operator $I_h : H^s(\Omega) \longrightarrow V_h$ by its restriction on any $T \in \mathcal{T}_h$. In what follows, we will be estimate the H^1 semi-norm and L^2-norm on Ω of $v - I_h v$. To do this, we need the following *a priori* error estimate of $v - I_T v$ on simplex T:

Lemma 4.1. *Let $T \in \mathcal{T}_h$. There exists a constant $c > 0$ independent of h, such that for all $v \in H^2(\Omega)$:*

$$|v - I_h v|_{1,T} \leqslant c\, d^2 h_T |v|_{2,T}, \|v - I_h v\|_{0,T} \leqslant c\, d^2 h_T^2 |v|_{2,T}. \tag{4.2}$$

Proof. The proof is analogous to the one given in [9], and it is based on the Bramble-Hilbert lemma (cf. [6]) and the passage to the reference simplex $\hat{T}$ of coordinates $(0,0)$, $(1,0)$ and $(0,1)$. As usually, let $F_T : \hat{T} \longrightarrow T$ be the affine transformation with the corresponding Jacobian matrix J_T, see [6]. Then, the following inequalities

$$|\hat{v}|_{m,\hat{T}} \leqslant c \|J_T\|^m |det(J_T)|^{-1/2} |v|_{m,T}, \quad \forall \hat{v} \in H^m(\hat{T}), \tag{4.3}$$

$$|v|_{m,T} \leqslant c \|J_T^{-1}\|^m |det(J_T)|^{1/2} |\hat{v}|_{m,\hat{T}}, \quad \forall v \in H^m(T) \tag{4.4}$$

hold, where

$$\|J_T^{-1}\| \leqslant \frac{h_{\hat{T}}}{\rho_T}, \quad \|J_T\| \leqslant \frac{h_T}{\rho_{\hat{T}}}, \quad |det(J_T)| = \frac{|T|}{|\hat{T}|}.$$

We first interest in the interpolation error in H^1 semi-norm on $\hat{T} = F_T^{-1}(T)$. We can write

$$|I_{\hat{T}} \hat{v}|_{1,\hat{T}} \leqslant \sum_{i=1}^{d+1} \frac{1}{meas(\hat{F}_i)} \left| \int_{\hat{F}_i} \hat{v} d\hat{s} \right| \hat{\varphi}_i|_{1,\hat{T}} \leqslant c \left(\sum_{i=1}^{d+1} |\hat{\varphi}_i|_{1,\hat{T}} \right) \|\hat{v}\|_{\mathcal{C}^0(\hat{T})}. \tag{4.5}$$

By using the Sobolev embedding theorem $H^2(\hat{T}) \subset C^0(\overline{\hat{T}})$, we show that there exists the constant $c > 0$ independent of $\hat{T}$ such that:

$$|I_{\hat{T}} \hat{v}|_{1,\hat{T}} \leqslant c \left(\sum_{i=1}^{d+1} |\hat{\varphi}_i|_{1,\hat{T}} \right) \|\tilde{v}\|_{2,\hat{T}}. \tag{4.6}$$

We can now apply the Bramble-Hilbert lemma to $\mathcal{F}(\hat{v}) = |\hat{v} - I_{\hat{T}} \hat{v}|_{1,\hat{T}}$, we have:

$$|\hat{v} - I_{\hat{T}} \hat{v}|_{1,\hat{T}} \leqslant c \left(\sum_{i=1}^{d+1} |\hat{\varphi}_i|_{1,\hat{T}} \right) |\hat{v}|_{2,\hat{T}}. \tag{4.7}$$

where the constant $c > 0$ is independent of $\hat{T}$.
By putting together (4.3), (4.4) and $\hat{I_T v} = I_{\hat{T}}\hat{v}$, we can get:

$$|v - I_T v|_{1,T} \leqslant c|\hat{v} - I_{\hat{T}}\hat{v}|_{1,T} \leqslant c\left(\sum_{i=1}^{d+1} |\varphi_i|_{1,T}\right) h_T |v|_{2,T}. \tag{4.8}$$

By injecting (2.23) of Proposition 2.7 in (4.8), we deduce existence of a constant $c > 0$ such that for any $T \in \mathcal{T}_h$ the following estimate holds:

$$|v - I_T v|_{1,T} \leqslant c d^2 h_T |v|_{2,T}. \tag{4.9}$$

In order to estimate the interpolation error in L^2 norm, we apply the same techniques and we use (2.23) from Proposition 2.7, which proves the announced result.

By summing each result of Lemma 2.4 over all simplex $T \in \mathcal{T}_h$, we get the estimate of the global interpolation error as follows.

Theorem 4.2. *(global interpolation error). There exists a constant $c > 0$ independent of h, such that for all $v \in H^2(\Omega)$:*

$$|v - I_h v|_{1,\Omega} \leqslant c\, d^2 h |v|_{2,\Omega}, \tag{4.10}$$

$$\|v - I_h v\|_{0,\Omega} \leqslant c\, d^2 h^2 |v|_{2,\Omega}. \tag{4.11}$$

5. Consistency error estimates

In this section we interest in the consistency error estimate for both two and three dimensions under considerations. To do this, we require the following lemma of Galerkin Orthogonality.

Lemma 5.1. *Let $u \in H^2(\Omega)$ be the exact solution of* (3.3)*, and u_h be the approximate solution of the variational formula* (3.6)*. Then, for all $v_h \in V_h$, the following Galerkin Orthogonality holds.*

$$a_h(u - u_h, v_h) = \sum_{F \in \mathcal{F}_h} \int_F \partial_n u [v_h]_F \, ds. \tag{5.1}$$

Proof. The proof is based on the integration by parts on Ω and on the use of relation $[xy]_F = [x]_F\{y\}_F + \{x\}_F[y]_F$. Then, we obtain

$$a_h(u, v_h) = l_h(v_h) + \sum_{F \in \mathcal{F}_h} \int_F [\partial_n u v_h]_F \, ds. \tag{5.2}$$

By putting together $[\partial_n u]_F = 0$ and $a_h(u_h, v_h) = l_h(v_h)$, we conclude with the desired result.

In the sense of (3.7), it is our aim to derive the consistency error estimate as follows.

Theorem 5.2. *(Consistency error). Let $u \in H^2(\Omega)$ be the exact solution of* (3.3)*. Then the consistency error estimate*

$$\sup_{w_h \in V_h,\, w_h \neq 0} \frac{a_h(u - u_h, w_h)}{\|w_h\|_{1,\Omega}} \leqslant c\, h |u|_{2,\Omega} \tag{5.3}$$

holds, with a constant $c > 0$ independent of h.

Proof. Estimate (5.3) is easy to proof by using the same steps in [9] and it is based on the Galerkin Orthogonality Lemma 5.1. One has for any $w_h \in V_h$ that

$$a_h(u - u_h, w_h) = \sum_{F \in \mathcal{F}_h} \int_F \partial_n u [w_h]_F \, ds. \tag{5.4}$$

By putting together, the Cauchy-Schwartz, the trace inequalities and the Lemma 2.4 for estimating the interpolation operator I_h, one has that:

$$\sum_{F \in \mathcal{F}_h} \int_F \partial_n u [w_h]_F \, ds = \sum_{F \in \mathcal{F}_h} \int_F \partial_n \left(u - I_h\right) [w_h]_F \, ds \tag{5.5}$$

$$\leqslant \left(\sum_{F \in \mathcal{F}_h} |F|^{-1} \|\partial_n (u - I_h u)\|_{0,F}^2 \right)^{1/2} \left(\sum_{F \in \mathcal{F}_h} |F| \|[w_h]\|_{0,F}^2 \right)^{1/2} \tag{5.6}$$

$$\leqslant c |u|_{2,\Omega} \left(\sum_{T \in \mathcal{T}_h} h_T^2 |w_h|_{1,T}^2 \right)^{1/2} \leqslant c\, h |u|_{2,\Omega} \|w_h\|_{1,\Omega}, \tag{5.7}$$

and the desired result follows readily.

We conclude from (3.7) that the *a priori* error estimate with respect to H^1-norm holds, with a constant $c > 0$ independent of h. In addition, by gathering the above result and the Aubin-Nitsch duality trick (cf. [2]), the $L2$-norm of *a priori* error holds, where a constant c is independent of h. So, we omit the details and we announce the following theorem:

Theorem 5.3. *Let $u \in H^2(\Omega)$ be the exact solution of* (3.3) *and let $u_h \in V_h$ be the approximate solution of the variational formula* (3.6)*. Then the a priori error estimate*

$$\|u - u_h\|_{1,\Omega} \leqslant c\, h |u|_{2,\Omega}. \tag{5.8}$$

holds, where $c > 0$ is independent of h.

If, in addition, Ω is convex and $f \in L^2(\Omega)$, then we have

$$\|u - u_h\|_{0,\Omega} \leqslant c\, h^2 |u|_{2,\Omega}. \tag{5.9}$$

6. Numerical experiments

To validate our proposed finite element method inspired by Crouzeix-Ravairt element method, different numerical experiments are investigated in two and three dimensions. Theoretical results for the Poisson's equation obtained from the *a priori* error analysis with respect to H^1-norm and L^2-norm, see Theorem 5.3, will be confirmed by several basic test cases by using the following Poisson model in $2D$ and $3D$:

$$\begin{cases} -\Delta u &= f \quad in\imath, \Omega, \\ u &= g \quad on\, \partial\Omega. \end{cases} \tag{6.1}$$

6.1. Convergence study in 2D

To confirm the optimal order *a priori* error estimate stated in Theorem 5.3, we study the error convergence for the two-dimensional Poisson's equation (6.1) in the unit square $\Omega = (0;1) \times (0;1)$ of $\mathbb{R}^2$. We choose the right-hand side f and the boundary condition g such that the exact solution u of (6.1) is given by:

$$u(x,y) = 12 + 11\left(x^2 + y^2 + xy\right) + 17x^3 + 20y^3. \tag{6.2}$$

For uniform mesh refinements, we present in Table 1 the computed errors, in H^1 and L^2 norms with the convergence rate calculated from the errors on two consecutive refined meshes, exactly with N and $4N$ triangles. From the presented table, we numerically obtain the optimal convergence rates (see Theorem 5.3) as follows:

$$\|u - u_h\|_{1,\Omega} = \mathcal{O}\left(N^{-1/2}\right) = \mathcal{O}\left(h\right), \quad \|u - u_h\|_{0,\Omega} = \mathcal{O}\left(N^{-1}\right) = \mathcal{O}\left(h^2\right). \tag{6.3}$$

N	$\|u - u_h\|_{1,\Omega}$	Order	$\|u - u_h\|_{0,\Omega}$	Order
8	1.3088e-00	——	9.8432e-01	——-
32	7.5127e-01	0.8008	3.0741e-01	1.6789
128	3.7164e-01	1.0154	7.9528e-02	1.9505
512	1.6983e-01	1.1298	2.0137e-02	1.9891
2048	7.2310e-02	1.2318	4.9275e-03	2.0309

Table 1: Errors and convergence orders.

In Figure 3, we represent these convergence orders in a $log - log$ scale. In Figure 4 from a qualitative point of view, we obtain the exact solution is very close to the discrete one, computed on the finest mesh with $N = 2084$ triangles.

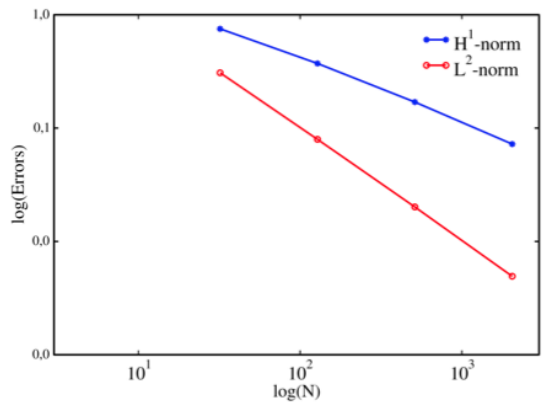

Figure 3: $2D$-Poisson model: Convergence rates of a finite element in $log - log$ scale.

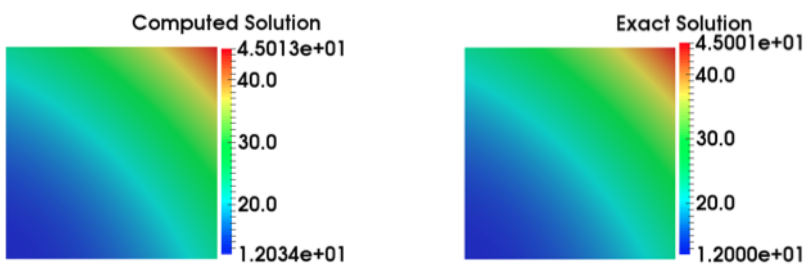

Figure 4: Approximate (left) and exact (right) solution of the Poisson's equation with our finite element in two dimensions.

6.2. Convergence study in 3D

In this subsection, we present our numerical results to support the theoretical analysis also in three dimensions. The Poisson's equation (6.1), is considered on the domain $(0;1) \times (0;1) \times (-1;1)$, where the volume force f is defined as

$$f(x, y, z) = 9\pi^3 \sin(\pi x) \sin(\pi y) \sin(\pi z) \tag{6.4}$$

and the Dirichlet boundary conditions are chosen such that the exact solution u of (6.1) is given by:

$$u(x, y, z) = 3\pi \sin(\pi x) \sin(\pi y) \sin(\pi z). \tag{6.5}$$

We show in Table 2 the errors computed in H^1 and L^2 norms as in the $2D$ example, but the convergence order is computed from the errors on two consecutive refined meshes with N and $4N$ tetrahedrons. The foregoing table show that the optimal orders are hold, that is $\mathcal{O}(h)$ for the H^1 norm and $\mathcal{O}(h^2)$ for the L^2 norm, which confirm the theoretical result in Theorem 5.3.

N	$\|u-u_h\|_{1,\Omega}$	Order	$\|u-u_h\|_{0,\Omega}$	Order
8	9.36576	——	4.96230	——-
32	5.28733	0.8248	1.37122	1.8555
128	2.81361	0.9101	0.36326	1.9164
512	1.37122	1.0369	0.08952	2.0207
2048	0.67971	1.0124	0.02191	2.0306

Table 2: Errors and convergence orders.

In what follows, the convergence orders of the approximation in $3D$ by the tetrahedral finite element are reported in Figure 5. In this case, we present also the exact solution and its computed. Upon the comparison of Figure 6, we obtain the exact solution is very close to the computed one, computed on the finest mesh with $N = 2084$ tetrahedrons.

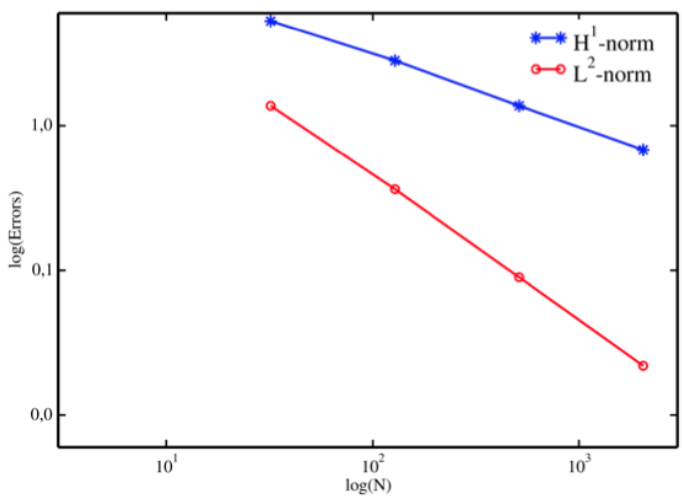

Figure 5: $3D$-Poisson model: Convergence rates of a finite element in $log - log$ scale.

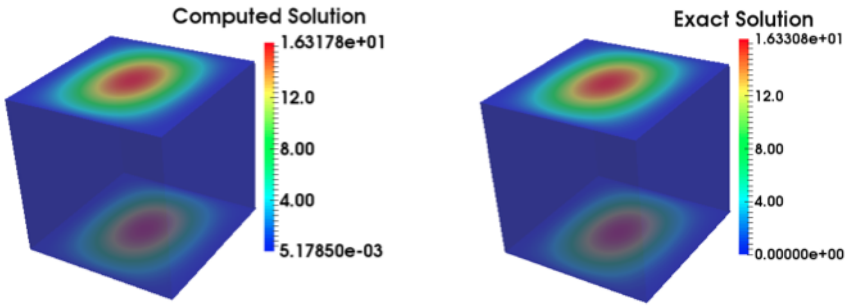

Figure 6: Approximate (left) and exact (right) solution of the Poisson's equation with our finite element in three dimensions.

7. Conclusion

In this paper, we have developed a virtual nonconforming finite element on triangle and tetrahedron and we applied it to Poisson's equation. We have proved that the finite element method yields an optimal *a priori* error estimate. To conclude, the numerical experiments in two and three dimensions are presented to validate the theoretical result. Future works are intended to be done in this field, such as the use of this finite element to prove the inf-sup condition for Stokes equation.

References

[1] D. Adak, E. Natarajan, S. Kumar, A new nonconforming finite element method for convection dominated diffusion-reaction equations. Int J Adv Eng Sci Appl Math 8, 274-283 (2016). https://doi.org/10.1007/s12572-016-0174-1.

[2] J.-P Aubin, Analyse fonctionnelle appliquée, Vols. 1 and 2. Presses Universitaires de France, Paris, 1987.

[3] M. Bachar, A. Guessab, A simple necessary and sufficient condition for the enrichment of the Crouzeix-Raviart element. Appl. Anal. Discrete Math. 10 (2016), no. 2, 378-393.

[4] G.A. Baker, Finite element methods for elliptic equations using nonconforming finite elements. Math. Comp. 1977, 31(137), 45-59.

[5] S. C. Brenner, R. L. Scott, The mathematical theory of finite element methods. Springer Science & Business Media, Vol. 15, 2007.

[6] P. G. Ciarlet, The finite element method for elliptic problems, North- Holland, Amsterdam, 1978.

[7] P. Ciarlet, Jr., C. F. Dunkl, S. A. Sauter, A Family of Crouzeix-Raviart Finite Elements in 3D. Analysis and Applications. 2017, 16, 10.1142/S0219530518500070.

[8] M. Crouzeix, P.-A. Raviart, Conforming and nonconforming finite element methods for solving the stationary Stokes equations. I. Revue Française d'automatique, informatique, recherche opérationnelle. Mathématique, 1973, 7(3): 33-75.

[9] A. Ern, J.-L. Guermond, Theory and practice of finite elements. Series: Applied Mathematical Sciences. Vol. 159, Springer, 2004.

[10] B.P. Lamichhane, A nonconforming finite element method for the Stokes equations using the Crouzeix-Raviart element for the velocity and the standard linear element for the pressure. Int. J. Numer. Meth. Fluids, (2014) 74: 222-228. doi:10.1002/fld.3848.

[11] R. Lim, D. Sheen, Nonconforming Finite Element Method Applied to the Driven Cavity Problem. Communications in Computational Physics, (2017) 21(4), 1012-1038. doi:10.4208/cicp.OA-2016-0039.

[12] R.S. Falk, Nonconforming finite element methods for the equations of linear elasticity, Math. Comp., 57 (1991) 529-550.

[13] J. Hu, R. Ma, The Enriched Crouzeix-Raviart Elements are Equivalent to the Raviart-Thomas Elements. J Sci Comput 63, 410-425 (2015), https://doi.org/10.1007/s10915-014-9899-9.

[14] K. Kobayashi, T. Tsuchiya, Error analysis of Crouzeix-Raviart and Raviart-Thomas finite element methods. Japan J. Indust. Appl. Math. 35, 1191-1211 (2018). https://doi.org/10.1007/s13160-018-0325-9.

[15] D. Shi, L. Pei, Nonconforming quadrilateral finite element method for a class of nonlinear sine-Gordon equation, Applied Mathematics and Computation 219 (2013) 9447-9460.

[16] D. Shi, L. Wang, X. Liao, Nonconforming finite element analysis for poison eigenvalue problem, Computers and Mathematics with Applications, Volume 70, Issue 5, 835-845, (Sept. 2015).

[17] D. Shi, H.Wang, The Crouzeix-Raviart type nonconforming finite element method for the nonstationary Navier-Stokes equations on anisotropic meshes. Acta Math. Appl. Sin. Engl. Ser. 30, 145-156 (2014), https://doi.org/10.1007/s10255-014-0274-2.